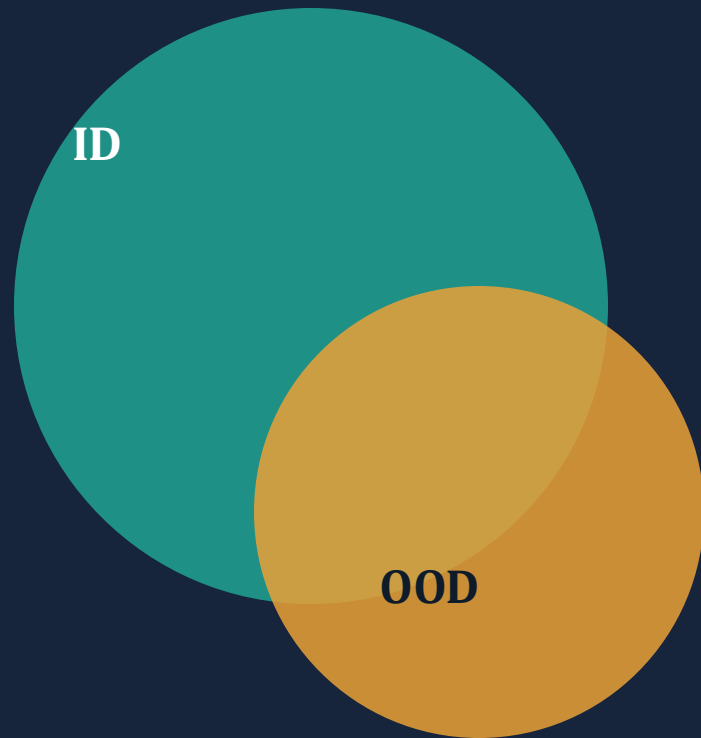


Computer Vision Project

OOD Detection with CORES for Monocular Depth Estimation

FastDepth and METER · NYU Depth V2 → KITTI

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Why OOD Detection Matters

1 The Problem

Outside the training domain, an MDE model produces plausible but unreliable depth maps, without signaling uncertainty.

2 The Idea

CORES, originally built for classification, uses convolutional kernel responses to build a score that separates ID and OOD data.

3 The Challenge

Dense prediction has no logits or classes: the monitored layers, spatial reduction, and decision threshold all need to be adapted.

Goal Are the internal convolutional responses of an MDE network enough to discriminate ID and OOD data? And how does this capability depend on the adaptation choices and the architecture?

Dataset: NYU Depth V2 and KITTI

	NYU Depth V2 (ID)	KITTI Depth Completion (OOD)
Scene	Indoor: offices, kitchens, bedrooms	Outdoor street scenes, from a moving car
Sensor	Microsoft Kinect (RGB + depth)	Velodyne LiDAR
Resolution	640×480	≈ 1242×375 (different aspect ratio)
Depth range	Up to ≈ 10 m	Up to ≈ 80 m
Depth map	Dense, with validity mask	Sparse: only LiDAR-hit points
Role	Training, calibration, ID test	OOD evaluation only

Shared preprocessing resize 224×224 · depth rescaled (10 m / 80 m) · ImageNet normalization

Four Distinct Data Splits

Training

85%

of the NYU training set

Calibration

15%

of the NYU training set:
threshold and scoring
hyperparameters

Test

ID

NYU test set: false-positive
evaluation

Test

OOD

KITTI: final evaluation only

- Calibration is never used for gradient updates
- Random split with a fixed seed, reproducible across runs
- ID test and KITTI are used only for final evaluation

The Two Models (from the Papers)

FastDepth

Lightweight CNN for real-time on embedded devices

Encoder: MobileNet (original V1; pretrained V2 in this project) with depthwise conv.

Decoder: 5 NNConv5 blocks (5×5 depthwise + 1×1 pointwise, nearest upsampling)

Additive skip connections

METER

Hybrid CNN-transformer for embedded devices (4 GB)

- Encoder derived from MobileViT: MobileNetV2 blocks + METER block (transformer on patches)
- Convolutional decoder with transposed convolutions and skip connections
- S, XS, XXS variants: 3.29M, 1.45M, 0.71M parameters

Evaluation Metrics

Depth Estimation

Metric	What It Measures
AbsRel	Mean relative error $ pred - gt / gt$
RMSE	Mean squared error, in meters
$\delta_1 \delta_2 \delta_3$	% of pixels with $\max(pred/gt, gt/pred) < 1.25 \cdot 1.25^2 \cdot 1.25^3$

OOD Detection

Metric	What It Measures
AUROC	$P(\text{ID score} > \text{OOD score})$; 0.5 = no discrimination
FPR95	False alarms on ID when 95% of OOD is detected
Acc. / Prec. / Rec. / F1	At the calibrated threshold, OOD = positive class
TNR	% of ID correctly classified as ID; expected ≈ 0.95
Balanced Acc.	Average of recall and TNR (290 ID vs 1000 OOD)

CORES in a Nutshell

1 Per-Kernel Extremes

For each kernel, only the positive max and negative min of the response map are kept.

2 Four Metrics

RM^+/RM^- (magnitude beyond the τ thresholds) and RF^+/RF^- (frequency of kernels beyond threshold). The four quantities are combined into the single-layer CORES score.

3 Multi-layer Score

The layer score is aggregated along a trajectory of kernels identified via backtracking.

Premise

ID Data

high peaks, deep troughs, many active kernels

OOD Data

muted responses, few active kernels

Structure of the Two Implementations

Input $3 \times 224 \times 224 \rightarrow$ output $1 \times 224 \times 224$ · BerHu, AdamW, OneCycleLR, 40 epochs · 10 monitoring points (E0–E4, D1–D5)

	FastDepth	METER
Stage 0–2	MobileNetV2, pretrained	Identical
Stage 3–4	MobileNetV2, pretrained	Inverted-residual + METERBlock, random weights
1×1 + BN projections (tap E0–E4)	64, 128, 256, 512, 1024 channels	64, 128, 256, 384, 512 channels
Decoder	5 NNConv5: upsample (nearest neighbor), 5×5 depthwise and 1x1 pointwise(each with BN and ReLu)	5 blocks: upsample(bilinear interpolation), 3×3 depthwise and 1x1 pointwise (each with BN and ReLu)
Decoder channels	512, 256, 128, 64, 32	384, 256, 128, 64, 32
Skip connection	Sum after pointwise conv's ReLU	Sum before pointwise conv's ReLU
Output	1×1 Conv + ReLU (raw tap)	Identical

Each encoder and decoder stage ends with a BatchNorm which is used as a Tap positioned before a ReLU

How the Score Is Computed

1 Hook

Single forward pass with hooks on 10 pre-ReLU BatchNorm taps (5 encoder + 5 decoder)

2 Spatial Reduction

Peak and minimum per channel on the H×W map

3 Four Metrics

RM^+ , RM^- , RF^+ , RF^- with per-layer τ thresholds

4 Layer Score

$S = \lambda_1 \cdot (\log RM^+ + \log RM^-) + \lambda_2 \cdot (\log RF^+ + \log RF^-)$
 $\lambda_1=2, \lambda_2=1, \varepsilon=1e-4$

5 Layer Averaging

One score per sample

6 Decision

Threshold = 5th percentile of ID validation; below threshold = OOD

Two Scoring Variants	Non-selected	Selected
Layers averaged	All 10	D1–D5 only
Channels used	All channels	Subsets from backtracking
RM^+ and RF^+	Computed on all channels	Computed on positive subset (far-region lineage)
RM^- and RF^-	Computed on all channels	Computed on negative subset (near-region lineage)
Paper equation	Eq. 5	Eq. 9

Backtracking Without Classes

Seed

- Pick the farthest 10% and nearest 10% of pixels of the raw output (analogues of `c_max` and `c_min`)
- Contribution defined by D5 channels activations weighted by the final 1×1 conv $\rightarrow$ pick top 20% of best contributors per region

Climbing from D5 to D1

- Take the pointwise 1×1 weight matrix connecting each current layer channels to previous layer channels (up5, up4, up3, up2). Extract the rows of the already-selected channels.
- Sum the weights across each column to determine influence value per channel of the previous layer.
- Pick 20% the channels with highest sum (TopK) and the channels with the lowest sum (BotK).

Channels Kept at Each Step

Step	Layer	Weights Used	FastDepth	METER
Seed	D5	Final 1×1 conv	Top 20%	Top 20%
1	D5 $\rightarrow$ D4	pointwise up5	13 of 64	13 of 64
2	D4 $\rightarrow$ D3	pointwise up4	26 of 128	26 of 128
3	D3 $\rightarrow$ D2	pointwise up3	51 of 256	51 of 256
4	D2 $\rightarrow$ D1	pointwise up2	102 of 512	77 of 384

- Stops at the decoder: skip connections are unweighted sums and the encoder is unreachable

Problems and Solutions (1/2)

Problem	Adopted Solution	Limitation
No classes or FC	c_max and c_min become the farthest and nearest depth regions	“Far/near” is not equivalent to “predicted class”
Backtracking	Decoder only (D5 → D1)	Skip connections are unweighted sums: the encoder is unreachable
Pre-ReLU taps	BatchNorm before ReLU to preserve RM^- and RF^-	Encoder stages have no clean tap
Encoder taps	Added 1×1 + BN projections, randomly initialized	Taps sit on a thin adapter, not on the backbone

Problems and Solutions (2/2)

Problem	Adopted Solution	Limitation
Transformer in METER	Taps read the projections only	Transformer responses are never evaluated directly
Numerical stability	Log-space, $\lambda_1=2$ (vs. 10 in the paper)	Departs from the original formula
Thresholds	Per-layer τ on noise, threshold at 5th percentile of ID validation	The selected score reuses the non-selected τ
Spatial reduction	Peak and minimum per channel on the H×W map	More than a single pixel extreme: τ needs recalibrating

Differences from the Papers

Aspect	FastDepth (paper → project)	METER (paper → project)
Encoder	MobileNet V1 → pretrained MobileNetV2, 320→1280 conv removed	MobileViT from scratch → 3 MobileNetV2 stages pretrained + 2 random METERBlocks
Transformer blocks	–	More blocks → only 2
Decoder	NNConv5 → NNConv5 with added projections	3 blocks with transposed conv → 5 blocks with bilinear upsample + depthwise
Loss & training	L1 → BerHu, AdamW, OneCycleLR	Balanced loss + shifting augmentation → BerHu, OneCycleLR, no augmentation
Size & input	–	S 3.29M, XS 1.45M, 256×192 → ≈2.5M, 224×224

Depth Quality and Calibration

1 Depth Quality

- ID test only (NYU): verifies the model is a good estimator
- AbsRel, RMSE, δ_1 – δ_3 on pixels with valid depth ($gt > 0.001$)
- Metrics computed per batch, then averaged

2 Calibrating the τ Thresholds

- Data: id_val + synthetic noise only
- id_val is passed through the model, for each channel of each layer the max and min value is kept
- Six percentile level candidates (50th–90th)
- For each layer and for each level, the thresholds τ_+ / τ_- are computed using the max/min distributions
- Scores on id_val and on 128 Gaussian-noise + 128 uniform-noise images
- Impose separate threshold :5th percentile of id_val scores. Noise images scoring above it are false positives
- The candidate with fewest false positives on noise is chosen

OOD Detection: Protocol

Three Disjoint Sets

Set	Role
id_val	Determines the decision threshold
id_test (NYU)	Measures behavior on ID data
ood_test (KITTI)	Measures behavior on OOD data

Evaluation

- Compute score. Higher score = more similar to ID data
- **Threshold-independent valuation** :
Compare score on ID test vs KITTI.
Metrics: AUROC and FPR95
- **Threshold based evaluation**:
Use 5th percentile of id_val on tests so that about 95% of ID is accepted.
Metrics: Accuracy, Precision, Recall, F1, TNR, Balanced accuracy

Threshold-independent valuation defines how good the score separates the ID from the OOD,

Threshold based evaluation defines how good the score is at detecting OOD at a specific point

The Four Experiments

Shared routine

Layer selection

Forward hook

Forward ID + OOD

CORES score

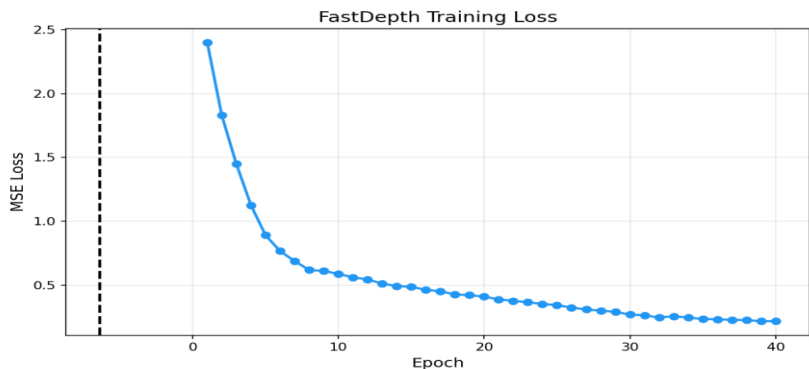
AUROC and FPR95

Experiment	Layer Subsets	Question It Answers
Per-layer	Each layer individually (10 runs)	Which layers are most discriminative?
Group	Encoder only, decoder only, both (3 runs)	Encoder or decoder: where is the information?
Cumulative	E0, then E0–E1, ... up to all 10 layers (10 runs)	How does performance change with depth?
Detailed components	A single run, all layers	Which component (RM^+ , RM^- , RF^+ , RF^-) separates best?

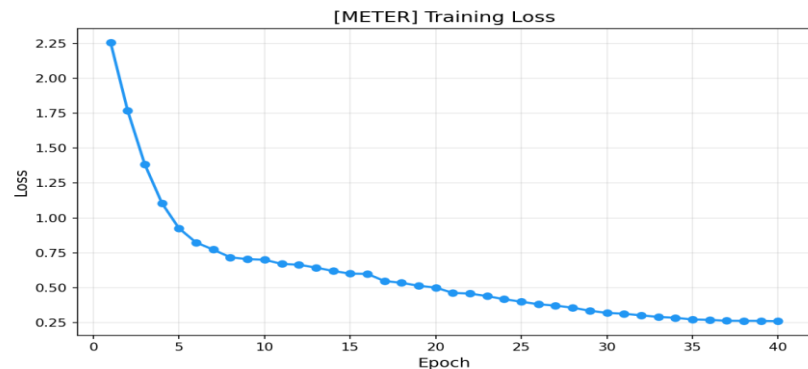
Detailed components is diagnostic: it does not compute AUROC, but records the mean RM^+ , RM^- , RF^+ , RF^- values for ID and OOD.

Behavior During Training

FastDepth (MobileNetV2 + NNConv5)



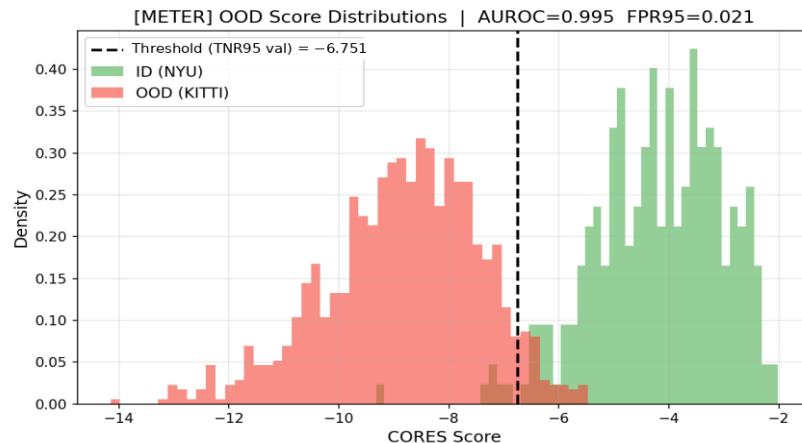
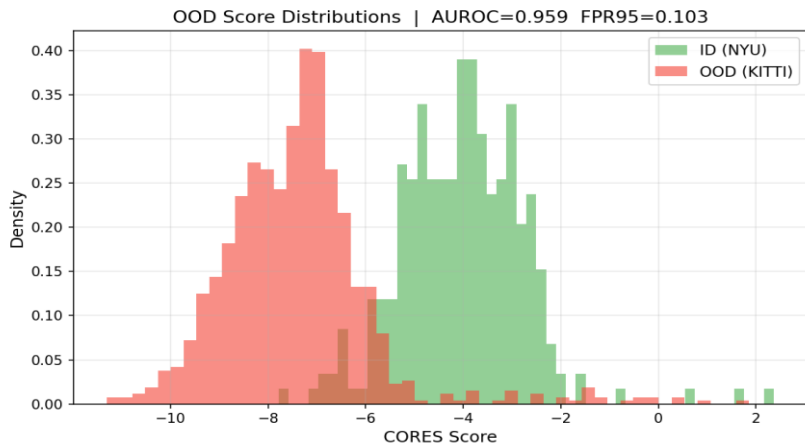
METER (MobileViT / Transformer + Conv Decoder)



Training behavior on NYU Depth V2 (40 epochs, BerHu loss, OneCycleLR):

- Fast, stable convergence: both models reach a plateau within the first 15–20 epochs, settling at a final loss of ~ 0.218 (FastDepth) and ~ 0.259 (METER).
- BerHu stability: combining L1 for small errors with adaptive L2 for large errors avoids gradient oscillations and produces well-calibrated dense predictions.

OOD Detection: FastDepth vs METER

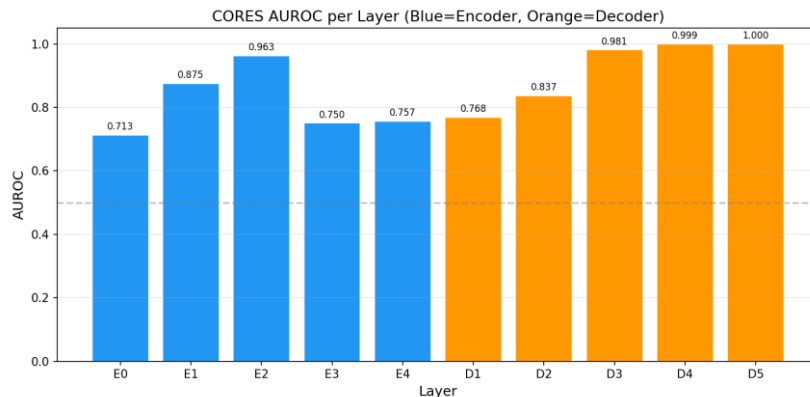
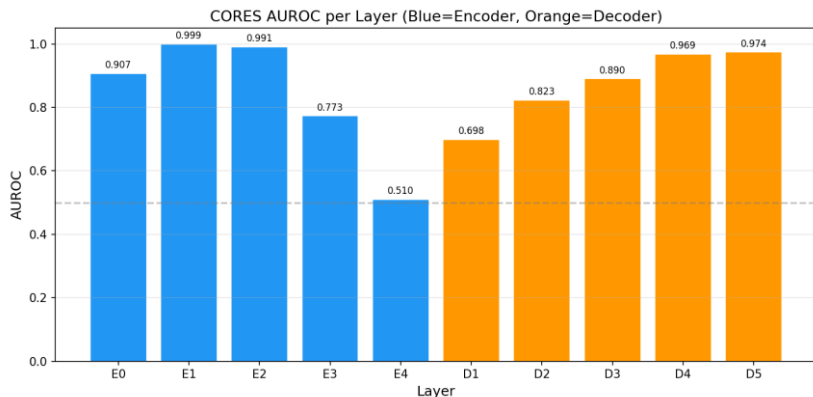


Performance Comparison | ID: NYU Depth V2 (290 test) vs OOD: KITTI Depth Completion (1000 test)

- FastDepth: AUROC = 0.959 · FPR95 = 10.3% · Calibrated threshold (TNR95 val) = -5.35
- METER: AUROC = 0.995 · FPR95 = 2.1% · Calibrated threshold (TNR95 val) = -6.75

Takeaway: Clear bimodal separation in both models. The hybrid CNN-transformer METER cuts FPR95 to a fifth compared to FastDepth, achieving near-perfect OOD discrimination (AUROC 0.995).

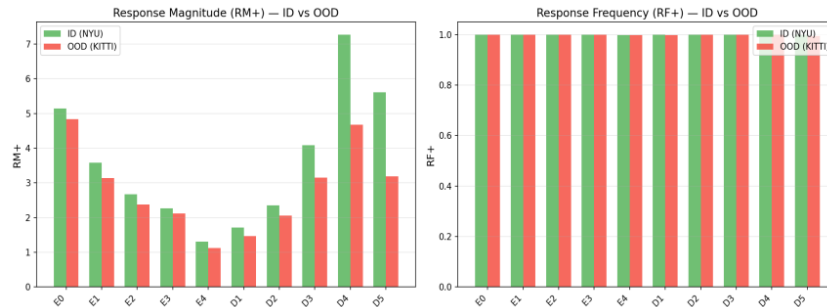
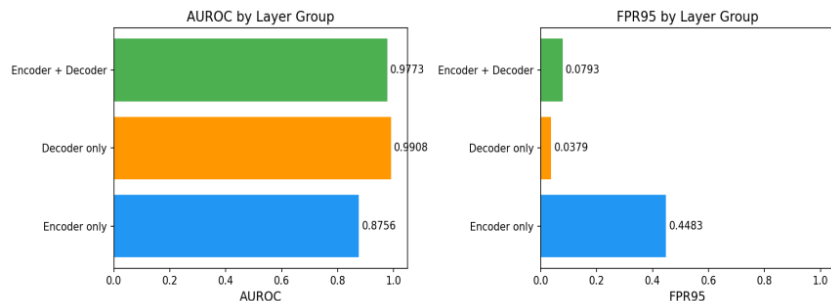
Ablation: Per-Layer Profile and U-Shaped Trend



Per-layer AUROC analysis (Blue = Encoder E0–E4, Orange = Decoder D1–D5):

- U-shaped trend: strong discrimination in the early layers (E0–E2, AUROC > 0.87–0.99) tied to low-level visual features (Kinect vs outdoor LiDAR sensor statistics).
- Bottleneck (E4): in FastDepth, AUROC collapses to 0.510 (chance level), while in METER the transformer blocks preserve a solid 0.757 thanks to global attention across patches.
- Recovery in the Decoder (D1–D5): discriminative power grows monotonically as the depth map is reconstructed, reaching 0.974 in FastDepth and 100% (1.000) in METER on D5.

Ablation: Layer Groups and CORES Components



1. Decoder-Only Beats All Layers

- In METER: Decoder-only reaches AUROC 0.9908 and FPR95 3.79%, surpassing the Encoder+Decoder combination (AUROC 0.9773, FPR95 7.93%).
- Same trend in FastDepth: Decoder-only (0.9525, FPR 9.3%) beats All (0.9430, FPR 16.5%). The encoder's deep layers “pollute” the averaged score.

2. RF+ Saturation and RM+ Dominance

- RF+ saturation: activation frequency (RF+) is 1.0 for both ID and OOD across all monitored layers; it offers no discriminative power.
- Separation driven by RM+: the entire OOD score is carried by magnitude (RM+), which in the decoder's final layers registers much stronger ID responses (7.3 vs 4.7 on D4).

Conclusions and Future Work

1. Effectiveness of CORES on Dense Prediction

Internal convolutional responses are sufficient to discriminate ID and OOD data even without logits or classes (AUROC > 0.95 in both models), confirming the initial research hypothesis.

2. CNN vs. Hybrid Transformer

METER outperforms FastDepth on every metric (AUROC 0.995 vs 0.959, FPR95 2.1% vs 10.3%). The transformer blocks prevent the specificity collapse at the bottleneck (0.757 vs 0.510).

3. Optimization: Monitor the Decoder Only

Limiting monitoring to the decoder improves discrimination (removes false alarms from intermediate layers) and halves the number of taps and the compute overhead — ideal for embedded/real-time deployment.

4. Future Work and Open Limitations

Extend backtracking through the additive skip connections to involve the encoder, and investigate transformer attention as a natural channel selector for the OOD score.